

ELEC70122: Machine Learning For Safety-Critical Decision-Making

Lecture 7: Causal Inference: Part 1 – Causes of Effects

Dr Sonali Parbhoo

AI for Actionable Impact (AI4AI), EEE

Imperial College London

s.parbhoo@imperial.ac.uk

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Credit: Several slides have been modified from Bauer, Ke (2022)

Outline

- Causality vs Correlation
- Structural Causal Models, Factorization and Identifiability
- Motivation for Causal Modelling and Challenges
- Deep Learning Methods for Learning Causes of Effects

Improving healthcare requires evidence

Randomised Control Trials (RCTs) are the gold standard for measuring treatment effects.



Treatment **A** or Treatment **B**?

- Subjects are **randomly assigned** each intervention.
- The effect of an intervention is measured as the difference between the outcomes of each group.

Figure adapted from ICML 2016 Tutorial by Shalit and Sontag.

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Randomised Control Trials as evidence?

Randomised Control Trials are **rare, biased and possibly wrong!**



Over 10% of 3,000+ of top journals studies are **“medical reversals”**.^[2]

[1] Smith, Saunders et al. Best Care at Lower Cost: The Path to Continuously Learning Healthcare in America, 2013

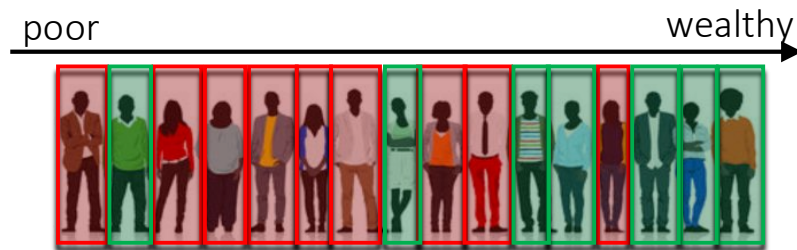
[2] Meta-Research: A comprehensive review of randomized clinical trials in three medical journals reveals 396 medical reversals. Herrera-Perez, Diana, et al. eLife 8 (2019): e45183 [https://elifesciences.org/articles/45183]

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Observational data as evidence?

- Observational data is not gathered to answer your hypothesis.
- Observational data suffers from **confounding**.



E.g. Socio-economic status is a confounder for treatment.

Can machine learning help us obtain evidence?

[1] Smith, Saunders et al. Best Care at Lower Cost: The Path to Continuously Learning Healthcare in America, 2013

[2] Meta-Research: A comprehensive review of randomized clinical trials in three medical journals reveals 396 medical reversals. Herrera-Perez, Diana, et al. eLife 8 (2019): e45183 [https://elifesciences.org/articles/45183]

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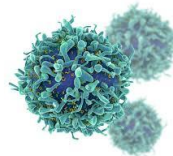
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Many questions in healthcare are causal

- Which type of treatment is best for me?



- Does gene X cause disease Y?



- Would my diagnosis be different, had my lifestyle habits changed?

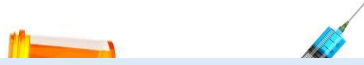


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Many questions in healthcare are causal

- Which type of treatment is best for me?

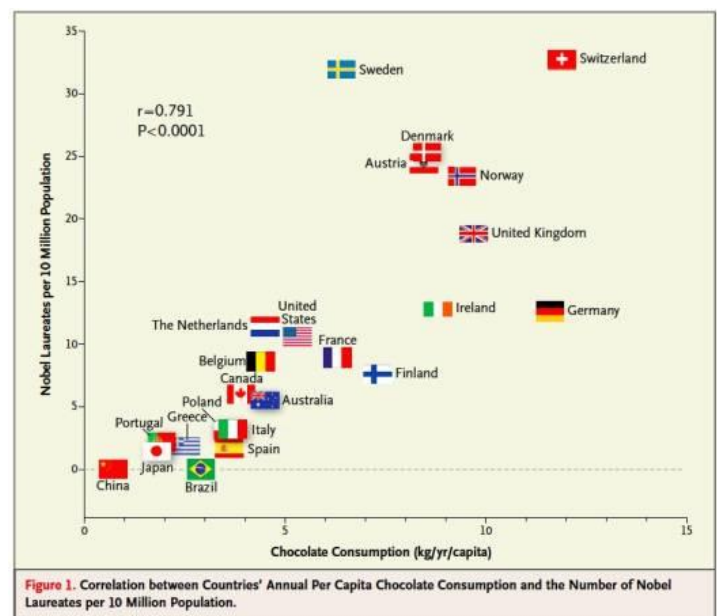
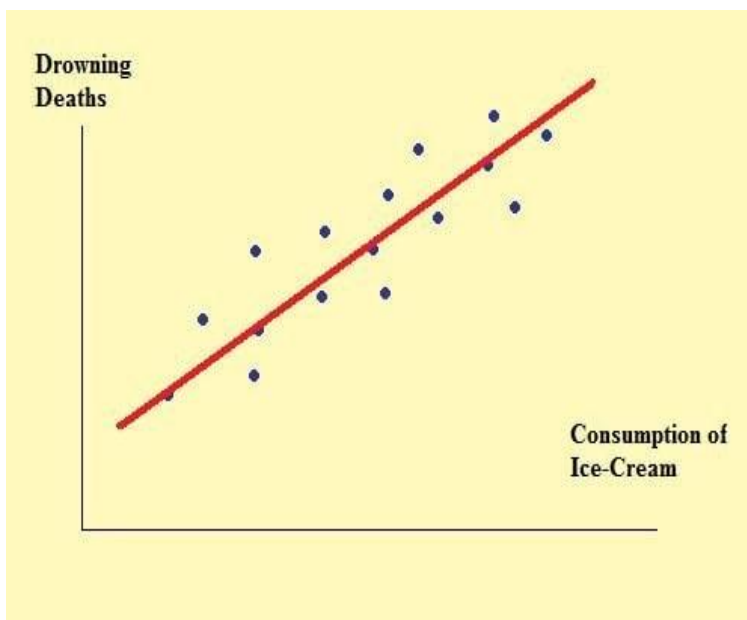


- Does getting a supervised learning 'hammer' to learn $p(y|x)$, but this is purely associational as it does not tell us what acting on x will **do** to y .

- Would my diagnosis be different, had my lifestyle habits changed?



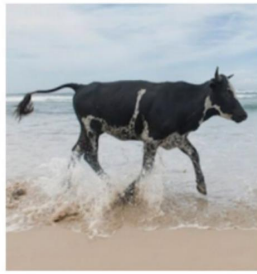
Many things are correlated!



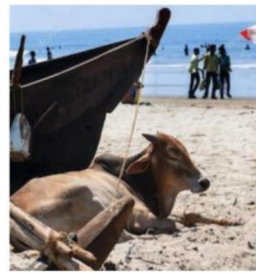
The problem with correlation: statistical vs causal dependency



(A) **Cow: 0.99**, Pasture: 0.99, Grass: 0.99, No Person: 0.98, Mammal: 0.98



(B) No Person: 0.99, Water: 0.98, Beach: 0.97, Outdoors: 0.97, Seashore: 0.97



(C) No Person: 0.97, **Mammal: 0.96**, Water: 0.94, Beach: 0.94, Two: 0.94

"Recognition in Terra Incognita."
ECCV 2018 Cows and grass are spuriously correlated.

Statistical Relationship (Correlation)

- The chances that grass and cow appears together.
- What most ML models capture

Causal Dependency

- Is the label "cow" actually caused by the cow in the image?

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Confounding Bias

- Does eating ice-cream cause death by drowning?
- Is something else causing both these phenomena?
- Could we realistically have some randomly chosen humans eat lots of ice-cream and see if what happens?
- In a healthcare setting, one cannot risk death because of the treatment!
- Outcomes are confounded by **season!**

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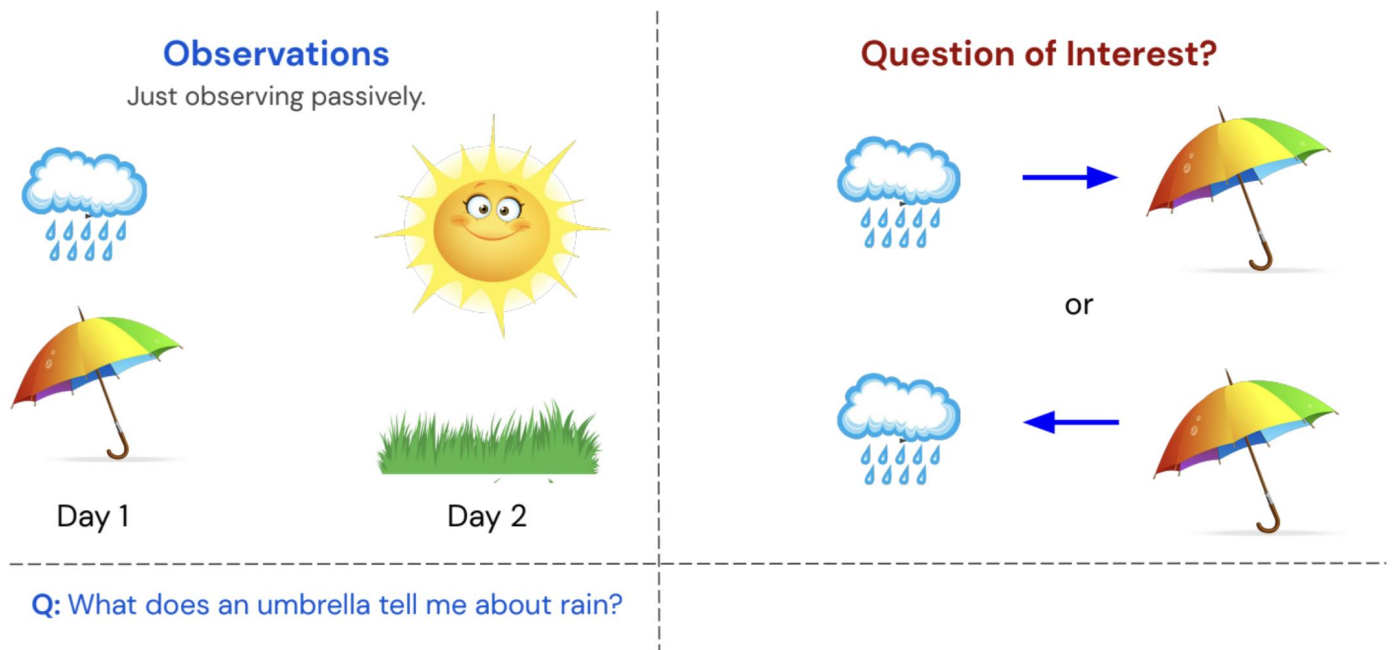
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Supervised learning is not enough

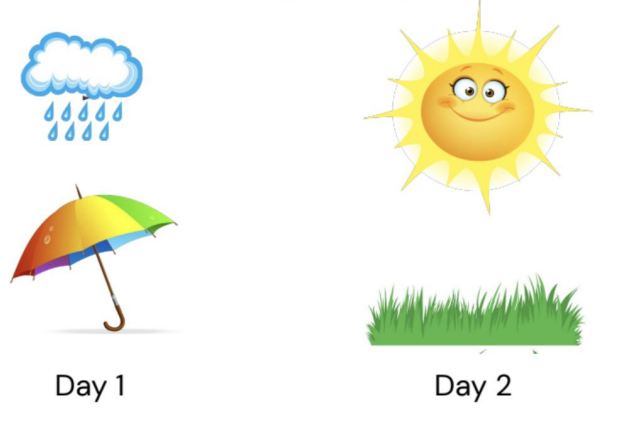
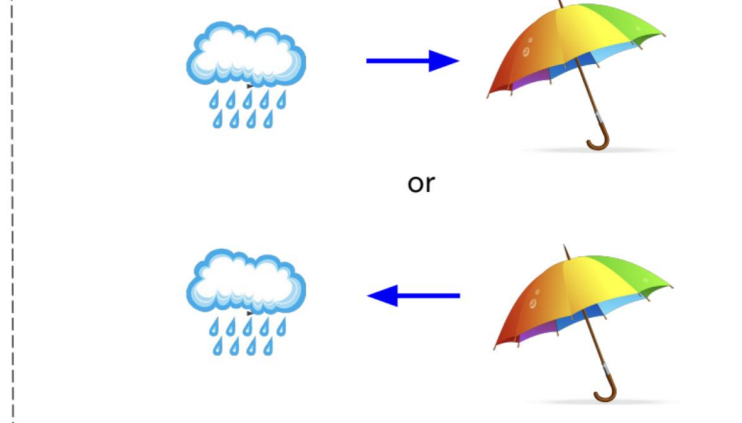


- A supervised model is optimized to predict outcome, not to differentiate between treatments **A** and **B**.
- Hidden factors may mean that **B** is worse than **A**, but rich patients usually take **B** and richer people also have better outcomes.

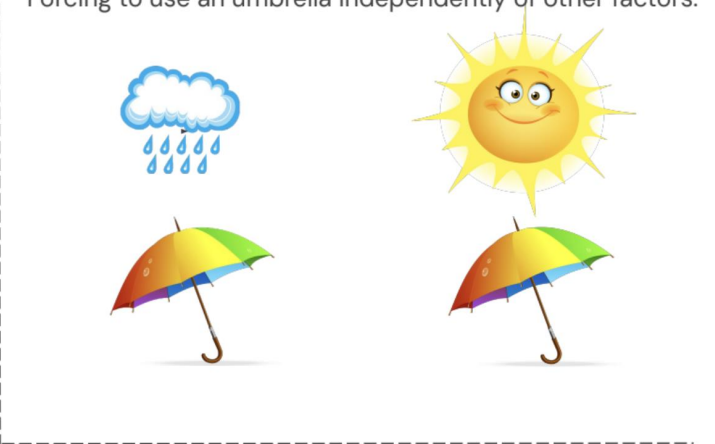
Identifying Causes and Effects



Identifying Causes and Effects

Observations Just observing passively.	Question of Interest?
 Day 1 Day 2	
Q: What does an umbrella tell me about rain?	A: Generally impossible from pure observations. → We need to make an experiment!

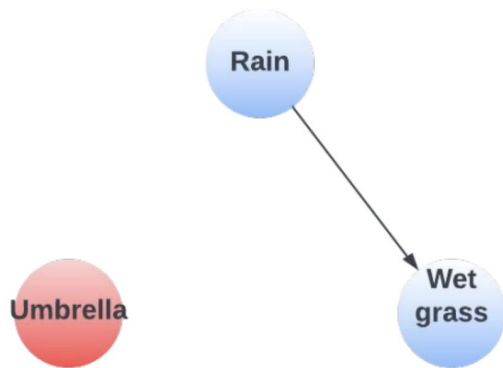
Identifying Causes and Effects

Observations Just observing passively.	Interventions Forcing to use an umbrella independently of other factors.
 Day 1 Day 2	
Q: What does an umbrella tell me about rain?	Q: What if I take an umbrella, will it rain? No!

Types of Interventions

Hard Intervention

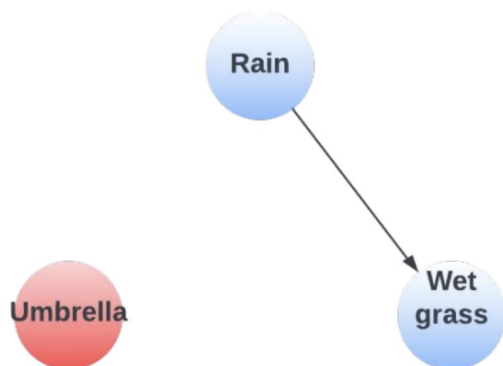
→ Causal parents no longer affect the child



Types of Interventions

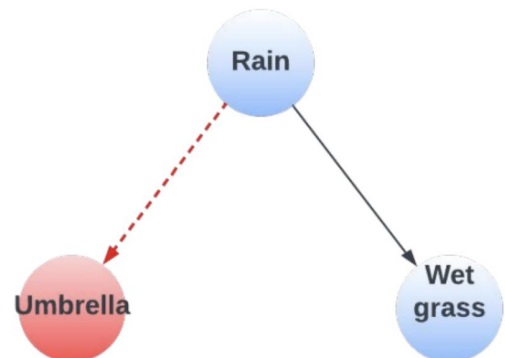
Hard Intervention

→ Causal parents no longer affect the child



Soft Intervention

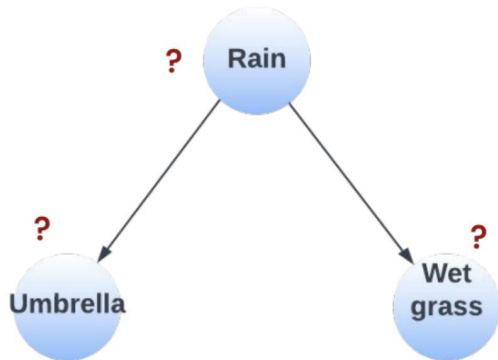
→ Changing how casual parents affect their child



Types of Interventions

Known / Unknown Intervention

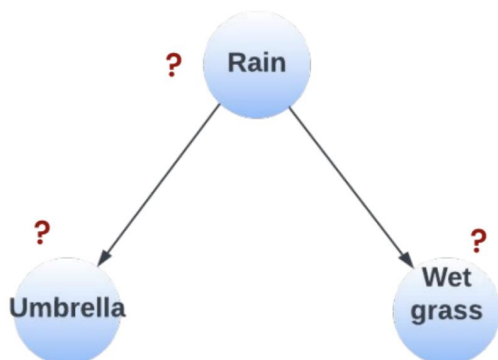
→ Do we know the target variable of the intervention?



Types of Interventions

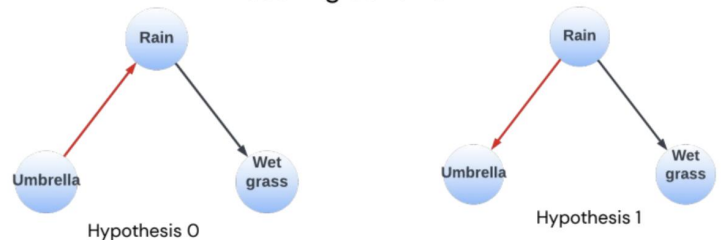
Known / Unknown Intervention

→ Do we know the target variable of the intervention?



Passive / Active Intervention

→ Active intervention is information seeking behavior.

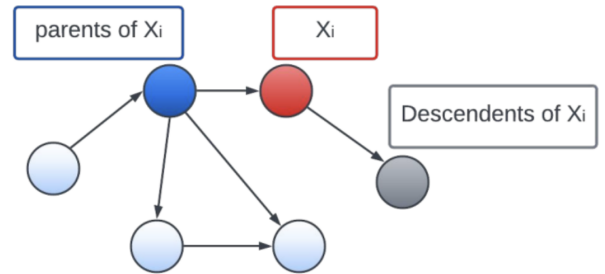


Active intervention:
To distinguish between hypothesis 0 and 1, intervening on "Umbrella" or "Rain" is more informative compared to intervening on "Wet grass".

Structural Causal Models

Describing the causal mechanisms of a system.

- **Variables:** $\mathbf{X} = \{X_1, \dots, X_N\}$
- **Noise:** $U = \{U_1, \dots, U_N\}$
- **Causal parents of X_i** $X_{pa(i, G)}$
- **Structural Equations:** $X_i = f_i(X_{pa(i, G)}, U_i) \quad \forall i \in \{1, \dots, N\}$
(Causal Mechanisms)

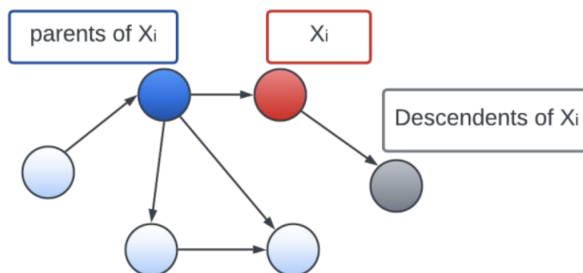


Causal Factorization

The arrows in SCM describes causal dependencies, rather than statistical correlations.

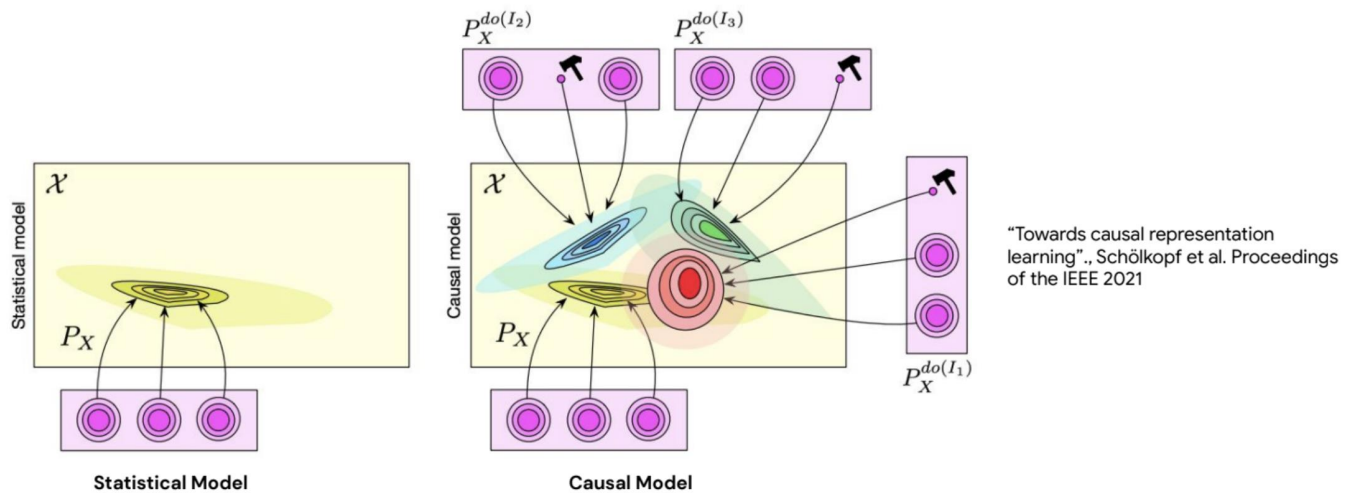
Hence, the causal factorization of the joint distribution is the following.

- **Factorization:** $P(X_1, \dots, X_N) = \prod_{i=1}^N P(X_i | X_{pa(i, G)})$



Causal models describes how intervening upon one variable affects others.

Differences between statistical and causal models



Causal models are more **powerful** than statistical models.

Every **intervention** defines a new **joint distribution**.

Statistical models can only capture one of them!

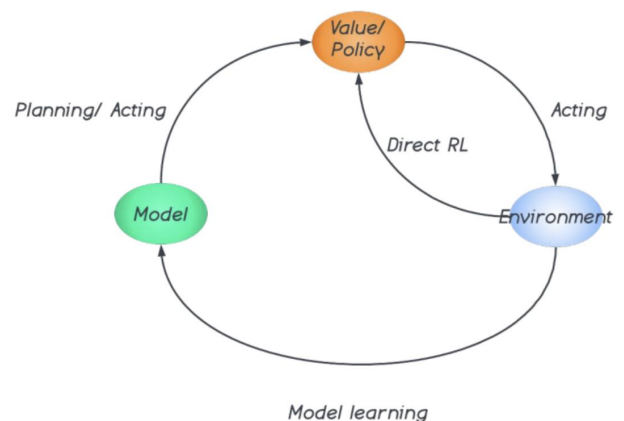
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The Need for Causal Modelling

Obtaining a **causal understanding of the environment** can help to improve:

- Acting
- Planning
- Credit Assignment
- Exploration



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The Need for Causal Modelling

Out-of-distribution generalization: Robustness to mismatches between train and test data.

Have causal graph →

- Know *that* colors don't affect physics of picking up objects
- Know *how* friction and mass affect picking up objects

∴ If robot has access to a causal model →

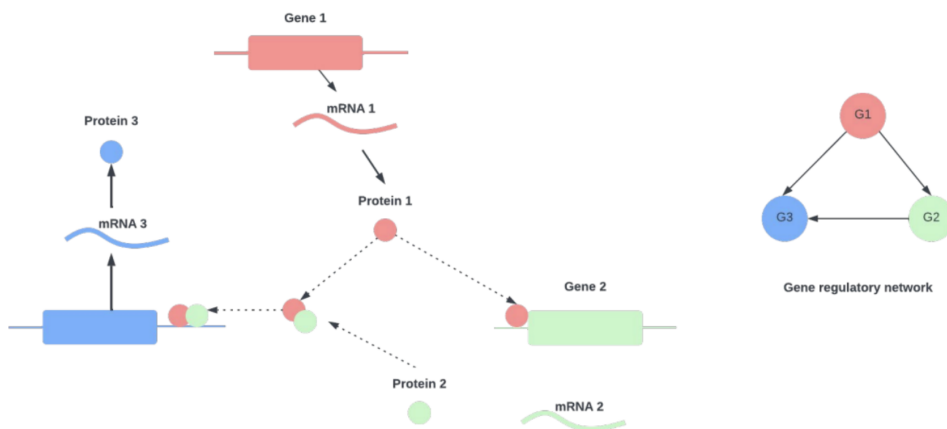
- Can generalize to previously-unseen
 - Colors
 - Shapes
 - Textures
 - Physical quantities



during test time.

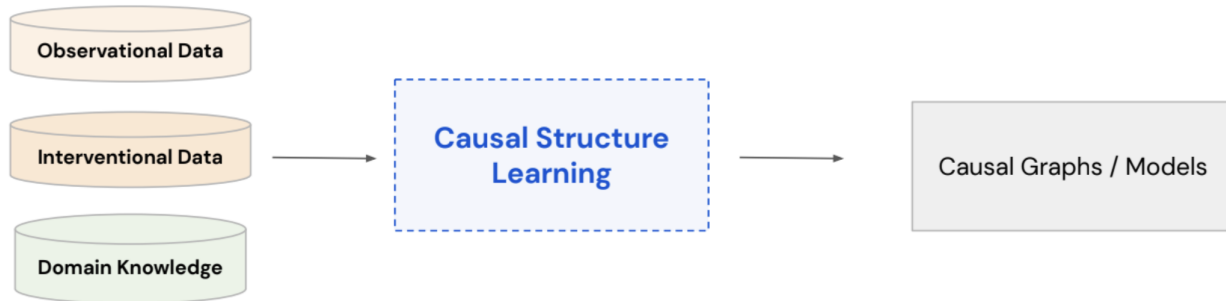
The Need for Causal Modelling

Gene regulatory networks: Understanding **causal relationships** between genes.



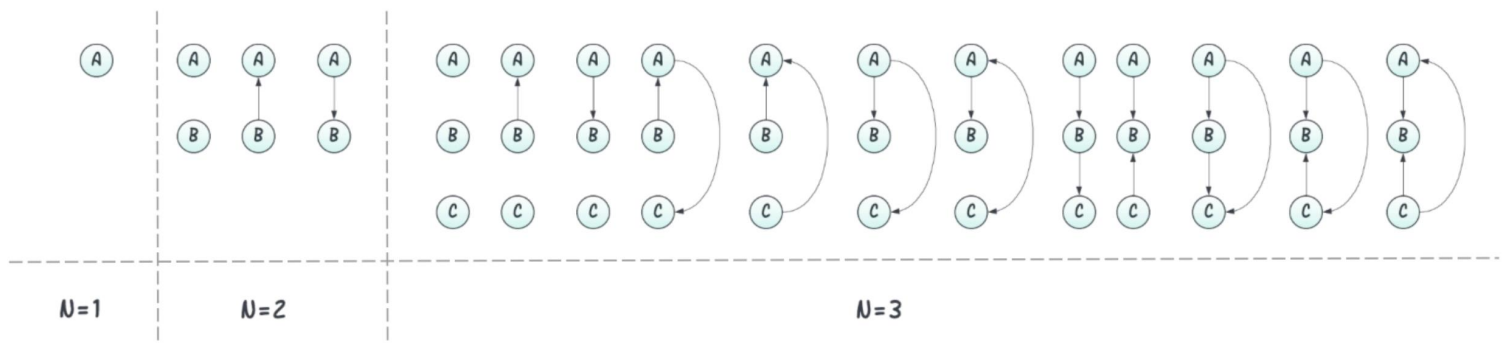
Causal Structure Learning: Causes from Effects

→ Inferring the underlying causal graphs from data and domain knowledge



Challenges of Causal Structure Learning

Space of causal graphs (i.e. DAGs) grows **super-exponentially**.



Identifiability

Identifiability: Given the data, can the ground-truth causal graph be discovered?

- Under pure observation, causal structure can only be identified up to Markov Equivalence Class.
 - Therefore no unique causal graph can be identified
- When:
 - Variables are given
 - Only the graph is to be learned
 - Exist sufficient **i.i.d. interventions**
- Then under certain assumptions, the graph can be **identified**.

MEC = Markov Equivalence Class
 G^* = Ground Truth Graph



Benefits and Limitations

Benefits

Going beyond Correlations

Reason about Cause and Effect

Capture Interventions

Limitations

Causal variables are given/known
[domain knowledge]

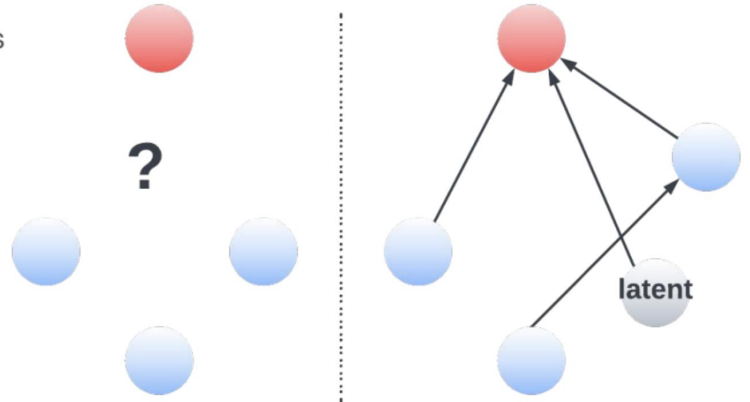
No learning of representations of
causal variables.

Scaling up can be challenging.

How do we learn causal graphs: Need for deep learning

Many real-world problems have **latent** variables.

- There might be many unobserved variables affecting climate change.
- Deep neural networks provide a highly expressive way of parameterizing latent variable models.



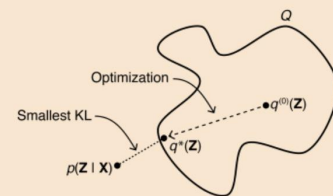
Deep Learning for Causal Graph Learning

A Modelling functional relationships (causal mechanisms)

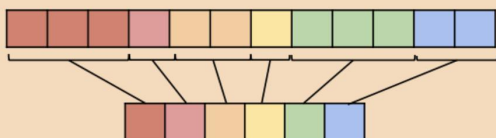
$$X_i = f_i(X_{pa(i, \mathbf{G})}, U_i)$$



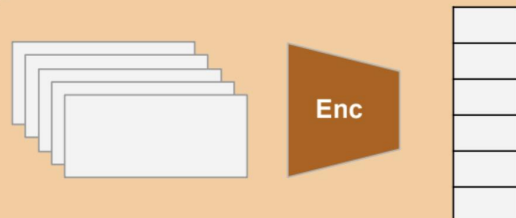
B Learn a distribution over graphs



C Learn representations of causal variables as rich compositions of learned features



D Learn latent causal variables



Causal Structure Learning

Goal: Learn a causal graph from data

Problems:

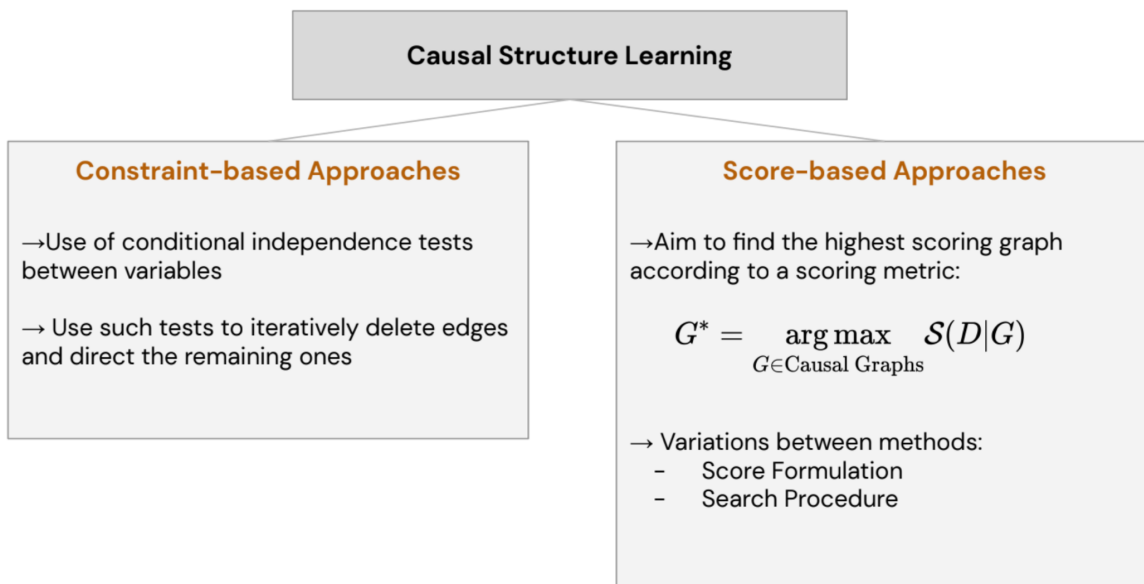
- Space of causal graphs grows super-exponentially
- Finding the right DAG is hard
- Limited identifiability from purely observational data

Challenges:

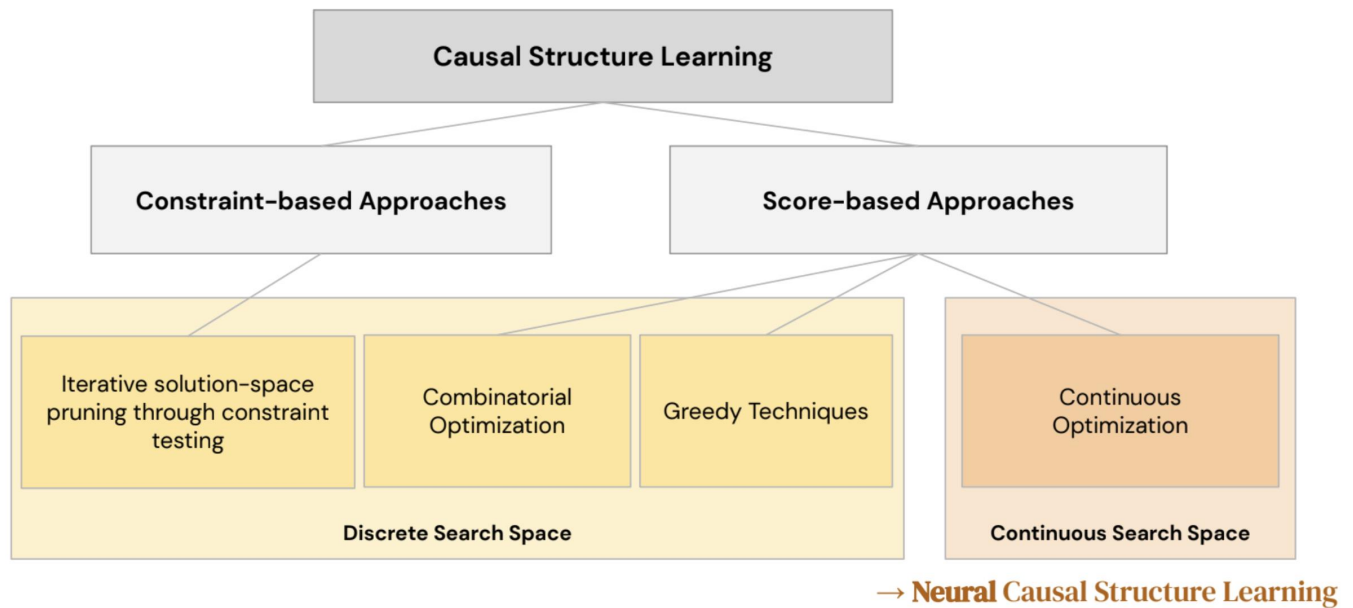
- Efficient Search in space of causal graphs

# of Nodes	# of DAGs	# of Graphs
1	1	1
2	3	2^2
3	25	2^6
4	543	2^{12}
5	29281	2^{20}
6	3781503	2^{30}
7	1138779265	2^{42}
8	783702329343	2^{56}
9	1213442454842881	2^{72}
10	4175098976430598143	2^{90}

Causal Structure Learning: Overview of Methods



Causal Structure Learning: Overview of Methods



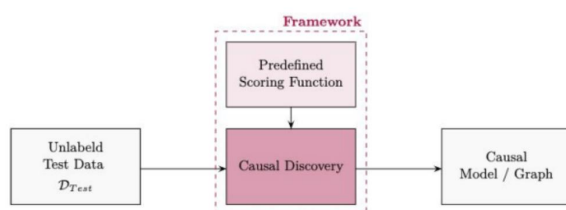
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Neural Causal Structure Learning: Paradigms

Unsupervised Frameworks

→ Infer a distribution of causal graph directly from test data using a predefined scoring function

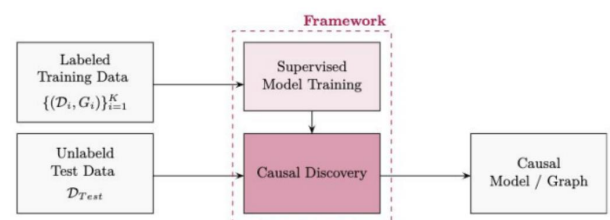


→ Currently the prevalent approach in literature

Supervised Frameworks

→ Training: Learn a mapping from datasets to graphs from labelled training pairs

→ Learn an amortized distribution

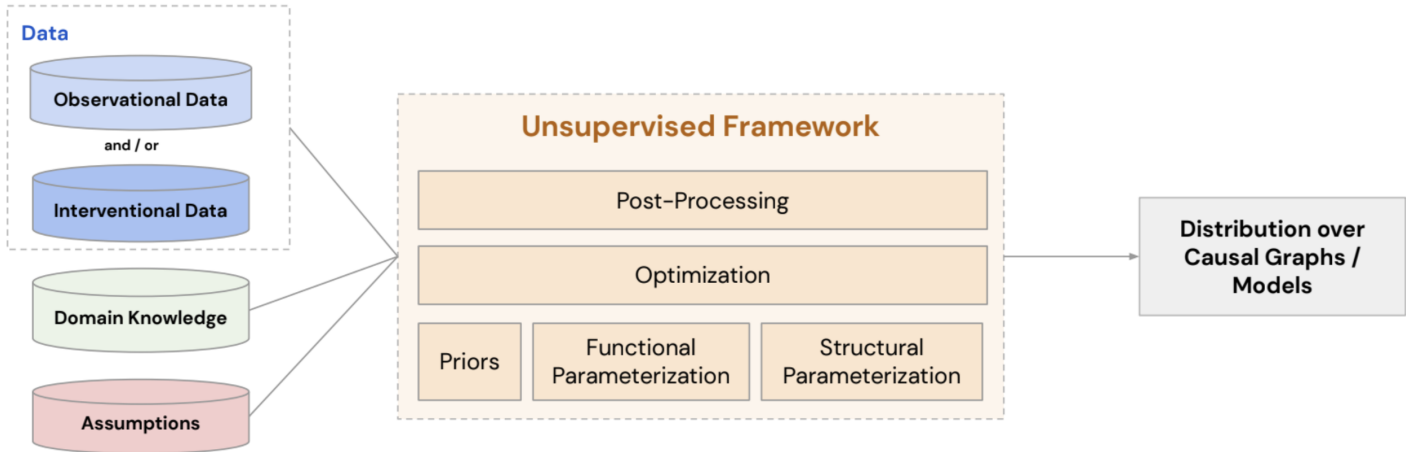


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Neural Causal Structure Learning: Unsupervised Methods

→ Infer a distribution over causal graphs directly from data using a predefined scoring function



Unsupervised Methods: Modelling Choices

Post-Processing:

How to refine an initial estimate of a causal graph?

Optimization:

How to learn the structural and functional parameters from data?

Functional Parameterization:

How to model functional relationships between variables?

$$X_i = f_i(X_{pa(i, \mathbf{G})}, U_i)$$

Structural Parameterization:

How to model the graph / distribution over causal graphs?

$$p(\mathbf{G}|\mathcal{D}) = \prod \dots$$

Priors:

How to incorporate prior knowledge and assumptions into the optimization framework?

Unsupervised Methods: Functional Parameterization

- **Q1:** How to model functional relationships between variables (i.e. **causal mechanisms**)?

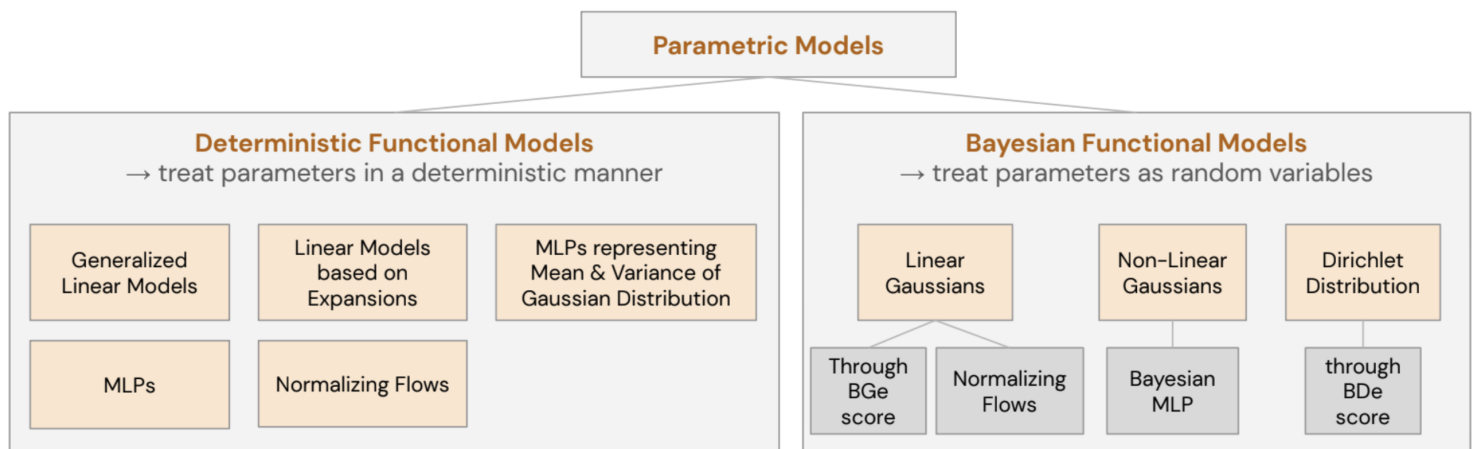
$$X_i = f_i(X_{pa(i, \mathbf{G})}, U_i)$$

- **Parametric models** (e.g. Generalized Linear Model, Neural Network, Normalizing Flow)
- **Non-Parametric models** (e.g. Gaussian Processes)

- **Q2:** How to treat the parameters of the **parametric functional models**?

- **Deterministic** Parametric Models
- **Bayesian** Parametric Models

Unsupervised Methods: Functional Parameterization



Unsupervised Methods: Structural Parameterization

Q: How to model the distribution over graphs?

- **Implicit Graph:** Model causal graph implicitly and infer it from functional parameters
→ Usually restricted to a point estimate of a graph
- **Explicit Distribution over Graphs:** Define a factorized distribution over graphs
 - Independent Factorization
→ assuming independence between edges
 - Non-Independent Factorization
→ model dependencies between edges

$$p(\mathbf{G}|\mathcal{D}) = \prod \dots$$

Unsupervised Methods: How to include prior knowledge?

Q: How to incorporate prior knowledge and assumptions into the optimization framework?

Functional Priors

- Enforced in functional parameterization (e.g. assumption of Gaussian data)
- Prior over parameters employed in **Bayesian Functional Models**

$$X_i = f_i(X_{pa(i, \mathbf{G})}, U_i)$$

Structural Priors

- Incorporation of Existing Knowledge / Belief (Edges, Subgraphs, ...)
- Acyclicity / DAGness Prior
- Sparsity Prior

$$p(\mathbf{G})$$

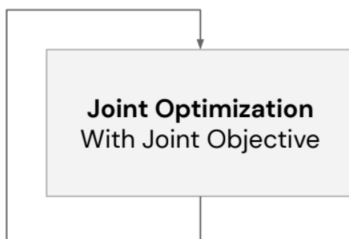
Unsupervised Methods: Optimization

Q: How to learn the structural and functional parameters from data in an unsupervised manner?

- **Optimization Structure**
 - **Joint Objective:** One Objective for Structural and Functional parameters
 - **Two Objectives:** Separate Objectives for Structural and Functional Parameters
- **Score**
 - **Scoring Function:** Score to optimize
- **Inference:**
 - **Bayesian Inference:** Treat all parameters in a Bayesian manner
 - **Classical Inference:** Treat certain parameters as deterministic parameters
- **Optimization:**
 - **Constrained Optimization**
 - **Unconstrained Optimization**

Unsupervised Methods: Optimization

(A) Joint Optimization:



(B) Optimization with Alternating Phases:



Functional Fitting:

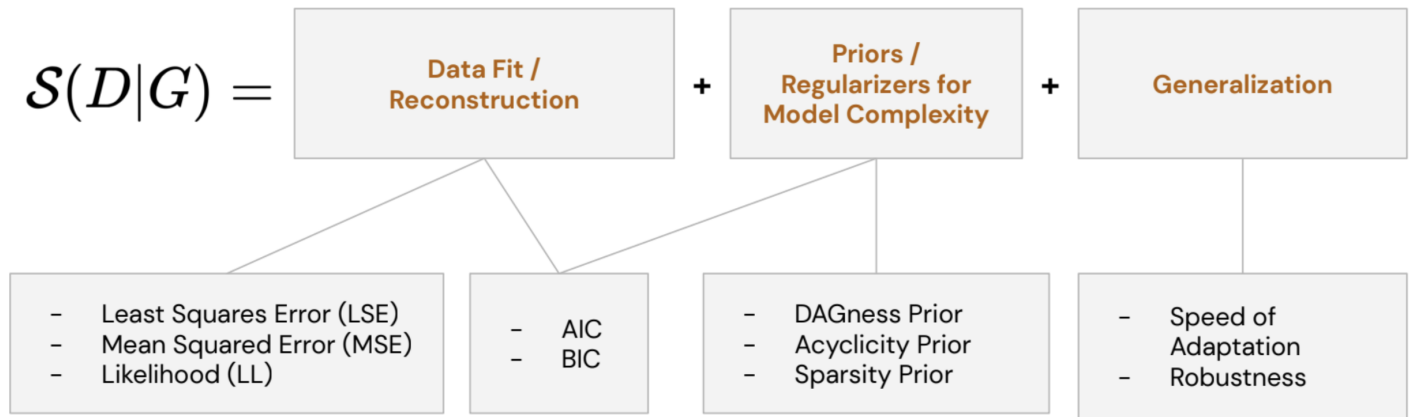
Optimize functional parameters under fixed structural belief

Structural Fitting:

Optimize structural parameters under fixed functional parameters

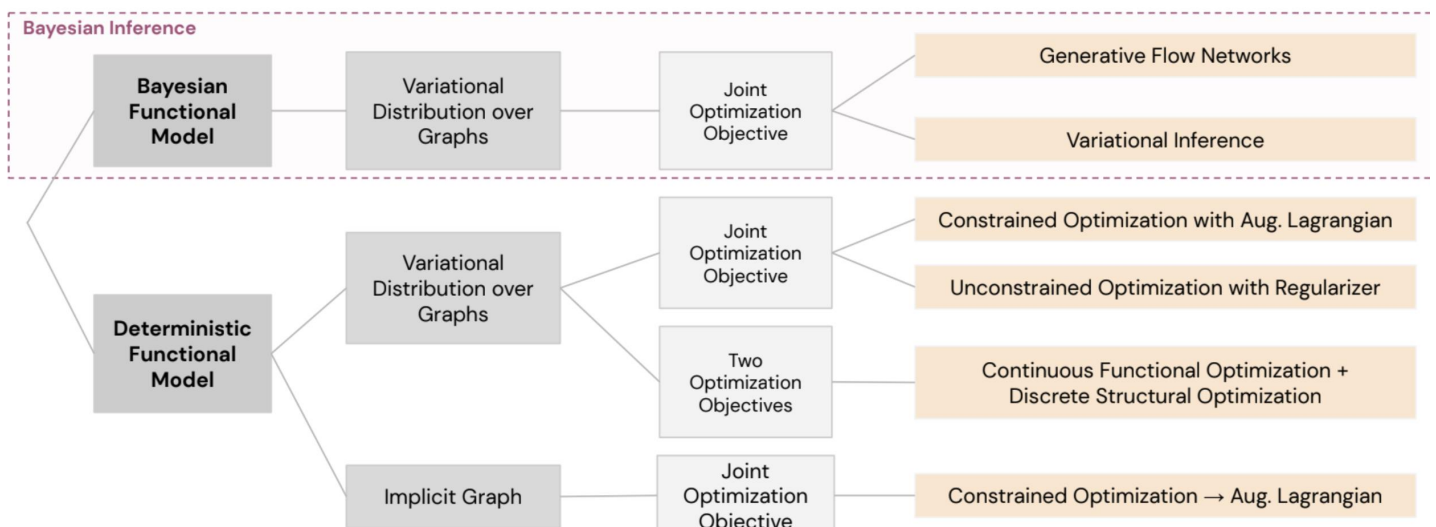
Unsupervised Methods: Optimization – Scoring Function

Find the graph G that optimizes the scoring function $\mathcal{S}(D|G)$:



Unsupervised Methods: Summary of Methods

→ Inference of **structural** and **functional** parameters

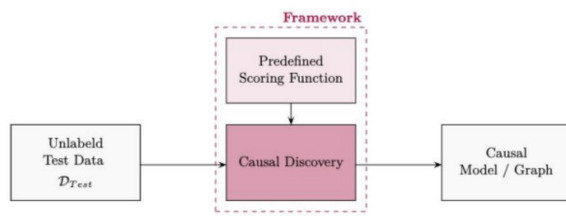


* The above classification summarizes employed techniques of existing approaches

Neural Causal Structure Learning: Paradigms

Unsupervised Frameworks

→ Infer a distribution of causal graph directly from test data using a predefined scoring function

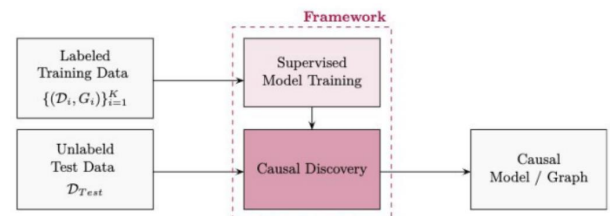


→ Currently the prevalent approach in literature

Supervised Frameworks

→ Training: Learn a mapping from datasets to graphs from labelled training pairs

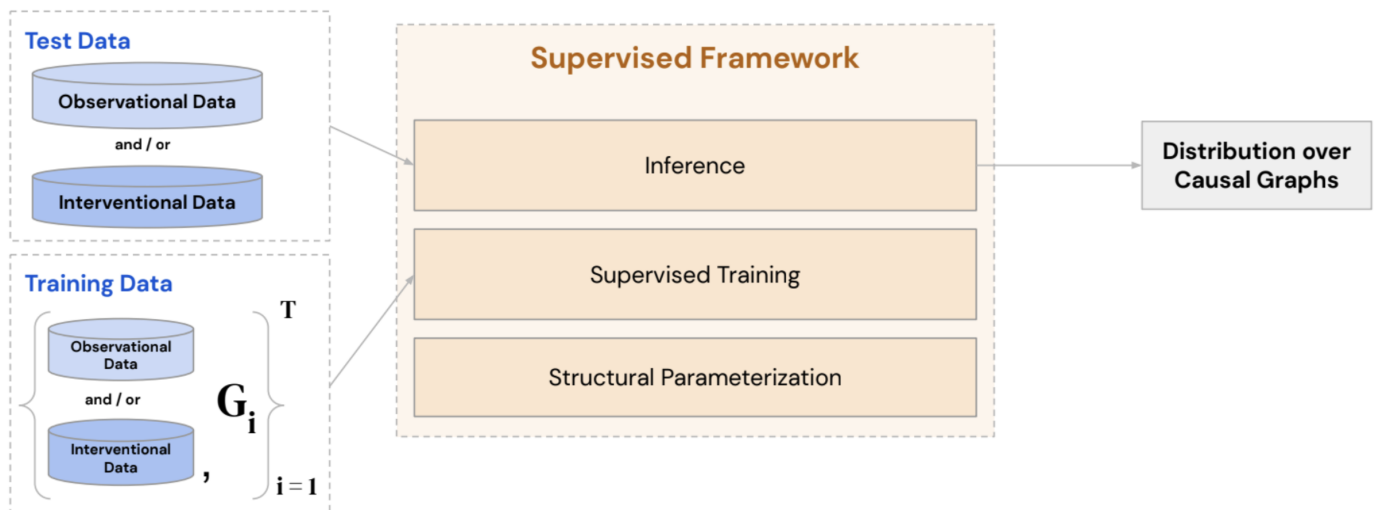
→ Learn an amortized distribution



Neural Causal Structure Learning: Supervised Methods

Training: Learn a mapping from datasets to graphs from labelled training pairs

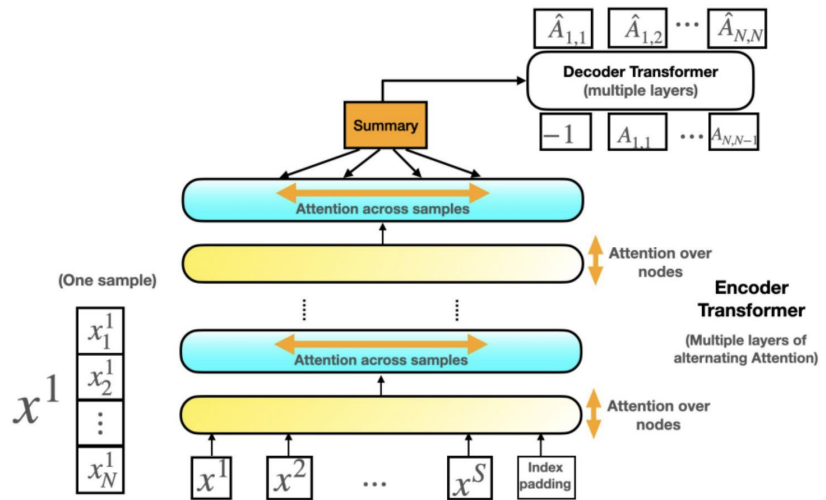
Inference: Infer distribution over graphs from test data



Supervised Methods: Transformer Models

Alternate attention in transformers

- Either learn **relationship between variables**.
- Or accumulate information learned across samples.

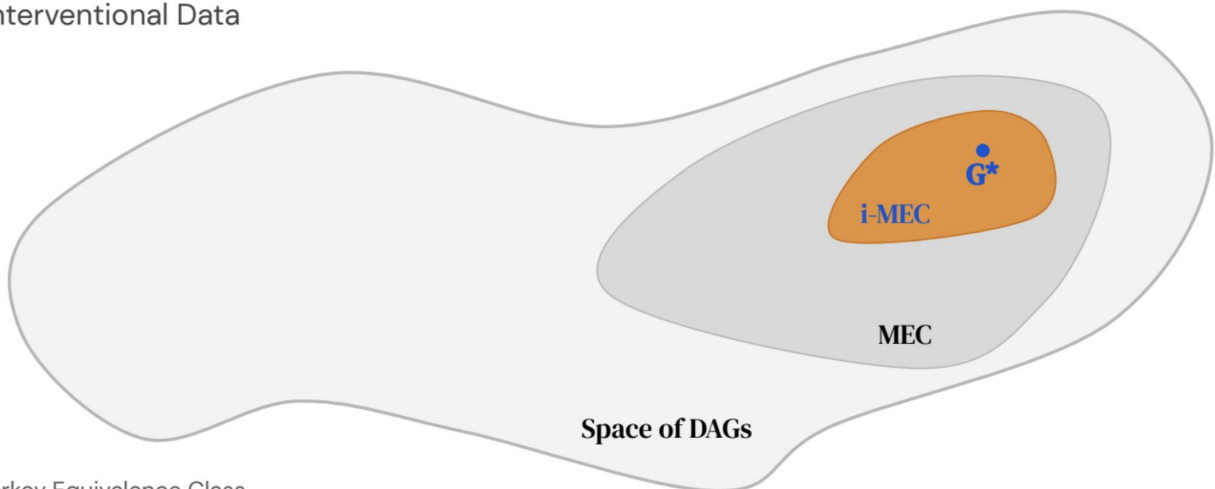


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How to improve identifiability?

1. Restrictive Assumptions on the functional nature of the data
2. Interventional Data



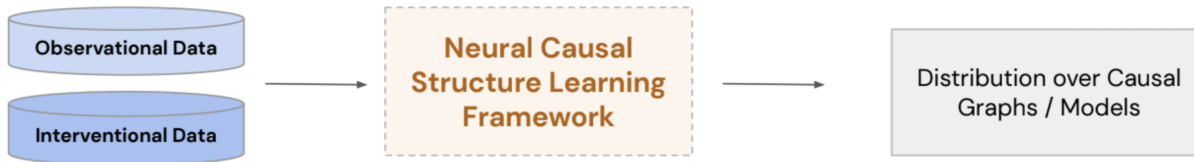
MEC = Markov Equivalence Class
i-MEC = Interventional Markov Equivalence Class
 G^* = Ground Truth Graph

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So far

→ **Static Procedure** = No active acquisition of data

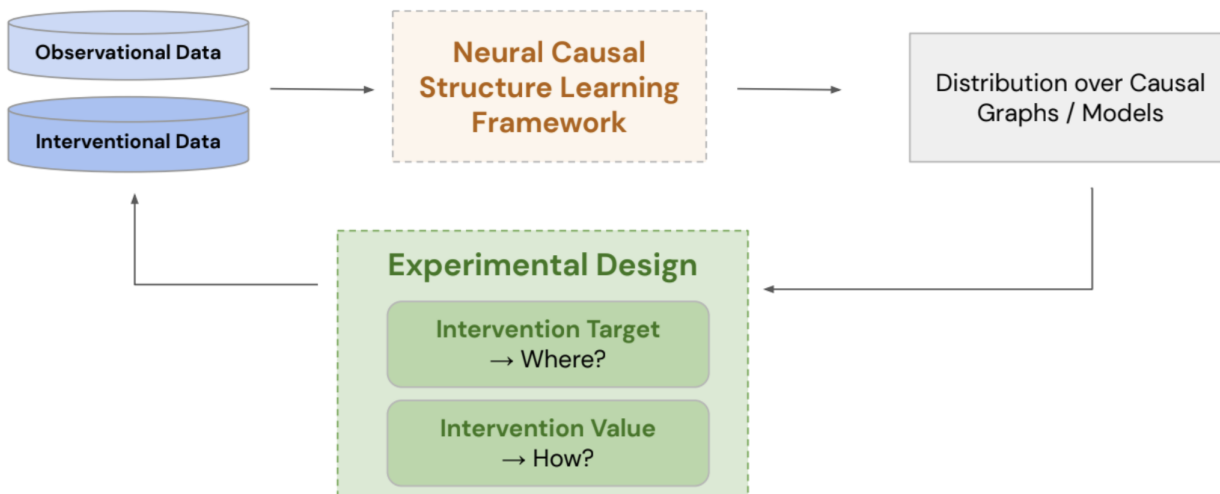


But the acquisition of interventional data tends to be:

- Expensive
- Technically impossible
- Sometimes unethical

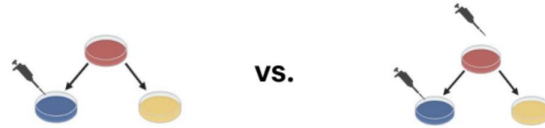
Experimental Design

→ Introduce a feedback loop to **actively** design experiments

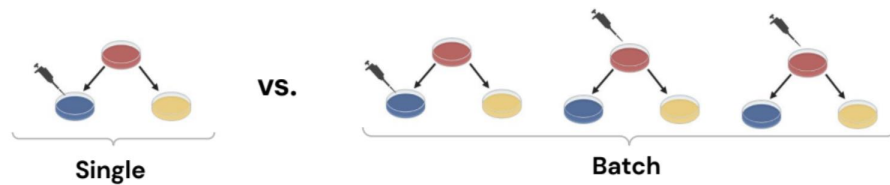


Experimental Design

- Number of Targets per Intervention
 - **Single Target** → perturb only one target in an intervention
 - **Multi Target** → perturb multiple targets in a single intervention



- Number of Interventions (Samples)
 - **Single Intervention** → Acquire a single intervention samples
 - **Batch of Interventions** → Acquire multiple intervention samples



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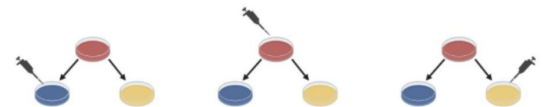
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Experimental Design

Q1: How to select the Intervention Target X_{Target} ?

- **Mutual Information based Approaches**
→ Maximizing expected information gain
- **Heuristic based Approaches**
→ E.g. disagreement between hypothetical samples
- **Diversity based Approaches**

WHERE TO INTERVENE?



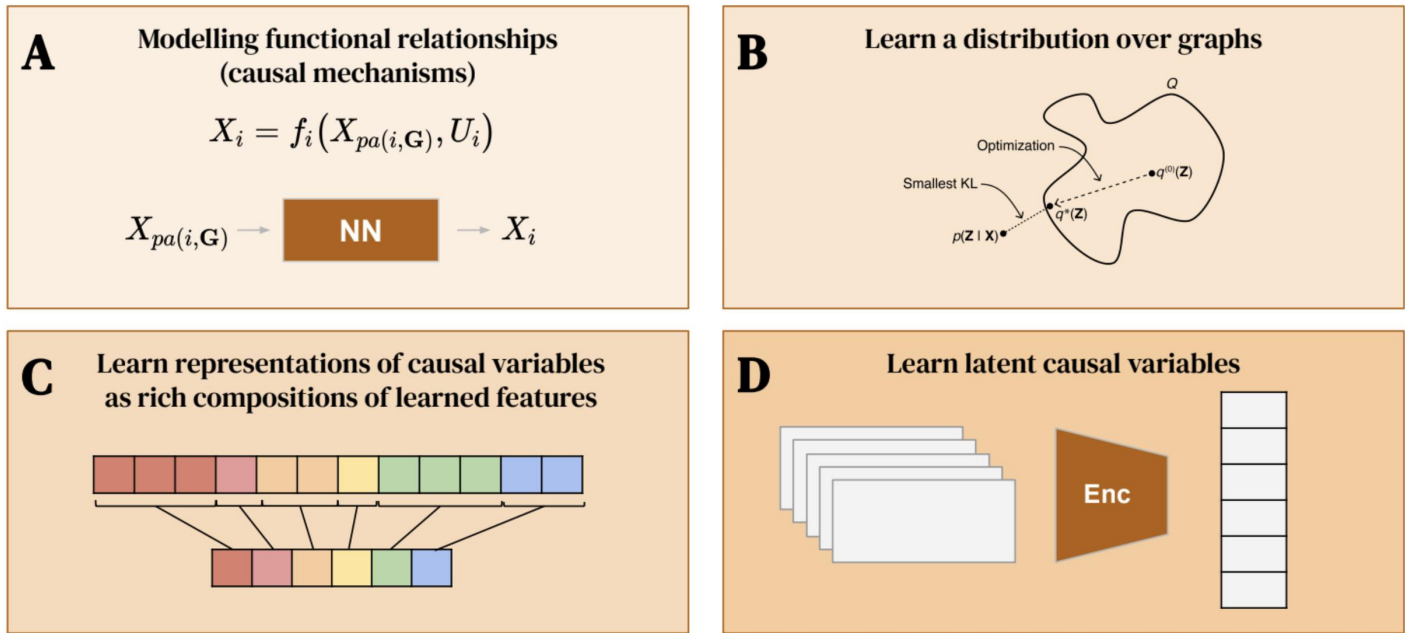
Q2: How to select the value(s) of the intervention $p(do(X_{Target}))$?

HOW TO INTERVENE?

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Deep Learning for Causal Graph Learning



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Causal Representation Learning as Feature Compositions

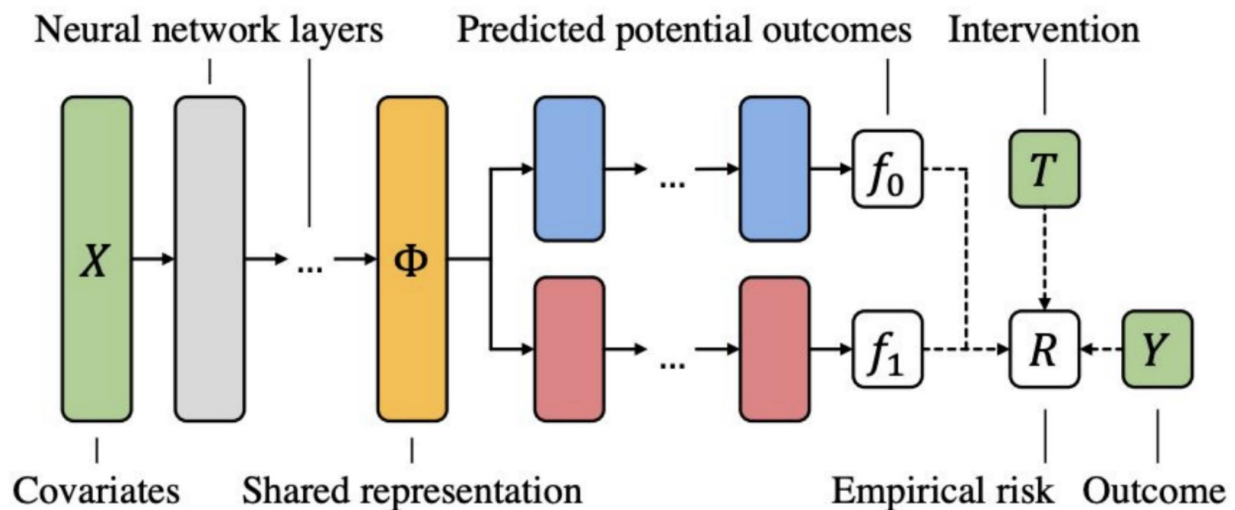


Image from: Johansson et al. "Generalization bounds and representation learning for estimation of potential outcomes and causal effects." *arXiv 2001.07426* (2020).

ELEC70122- Lecture 7: Causal Inference

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Summary and Open Topics

- Deep Learning for Causality has significant applications beyond structure search!
- Relatively few works on Bayesian Causal Discovery and supervised settings
- Concrete Downstream Tasks (e.g. Missingness, Experimental Design, Cause-Effect Estimation) rather than evaluation based on metrics
- Sample Efficiency with respect to both observational and interventional data sets.
- High-dimensional setting as way towards Representation Learning
- Time Series is yet unsolved despite time being very helpful for inferring causal structure.
- Real-World datasets often lack groundtruth: Need efficient simulators!

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